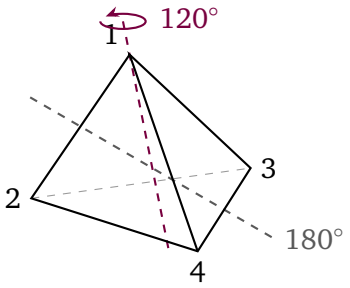
Symmetries in Physics — Exercise Sheet 5

Characters and character tables

Variant: Questions only

Exercise 1. Construct the character table of A_4, the rotation group of the tetrahedron. The picture, taken from Lecture 2, shows one rotation axis of each kind:



Here is everything you will need (Lecture 2's example $A_4/V \cong C_3$).

- $|A_4| = 12$, and A_4 has four conjugacy classes: the identity; the three half-turns $(1\,2)(3\,4)$, $(1\,3)(2\,4)$, $(1\,4)(2\,3)$; the four 3-cycles $(1\,2\,3)$, $(1\,3\,4)$, $(1\,4\,2)$, $(2\,4\,3)$; and the four inverse 3-cycles $(1\,3\,2)$, $(1\,4\,3)$, $(1\,2\,4)$, $(2\,3\,4)$.
- The identity and the three half-turns form a normal subgroup V, and $A_4/V \cong C_3$. The projection $\pi\colon A_4 \to C_3$ sends V to e, the class of 3-cycles containing $(1\,2\,3)$ to r, and the class of $(1\,3\,2)$ to r^2.
- The characters of C_3 are $\chi_k(r^m) = \omega^{km}$, with $\omega = e^{2\pi \mathrm{i}/3}$ and $k = 0, 1, 2$.

(i) Without constructing any representation, use the two counting theorems of Lecture 5 to find the dimensions of the irreducible representations of A_4. Do the same for D_6, which has order 12 and six conjugacy classes (Sheet 2, Exercise 2).

(ii) Construct three one-dimensional representations of A_4 as $\chi_k \circ \pi$.

(iii) Find the missing row of the table from $\sum_i n_i \chi_i(g) = 0$ for $g \neq e$, and check its norm.

Exercise 2. Lecture 5's example on D_4 claimed that the quaternion group Q_8 has the same character table. Prove it. Here is everything you will need (Sheet 4, Exercise 6).

- $Q_8 = \{\pm 1, \pm \mathrm{i}, \pm \mathrm{j}, \pm \mathrm{k}\}$, with $\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = \mathrm{ijk} = -1$. Any two of $\mathrm{i}, \mathrm{j}, \mathrm{k}$ anticommute, for instance $\mathrm{ij} = \mathrm{k} = -\mathrm{ji}$.
- Four one-dimensional representations ψ_{ab}, with $a, b = \pm 1$:

$$\psi_{ab}(\pm 1) = 1, \quad \psi_{ab}(\pm \mathrm{i}) = a, \quad \psi_{ab}(\pm \mathrm{j}) = b, \quad \psi_{ab}(\pm \mathrm{k}) = ab.$$

- A two-dimensional irreducible representation

$$\mathrm{i} \mapsto -\mathrm{i}\sigma_1, \quad \mathrm{j} \mapsto -\mathrm{i}\sigma_2, \quad \mathrm{k} \mapsto -\mathrm{i}\sigma_3, \quad -1 \mapsto -\mathbb{1}.$$

Here $\sigma_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma_2 = \left(\begin{smallmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{smallmatrix}\right)$, $\sigma_3 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ are the Pauli matrices.

- The table of D_4 (Lecture 5):

D_4	$1\,e$	$1\,r^2$	$2\,r$	$2\,s$	$2\,rs$
A_1	1	1	1	1	1
A_2	1	1	1	−1	−1
B_1	1	1	−1	1	−1
B_2	1	1	−1	−1	1
E	2	−2	0	0	0

  The lecture labels these rows $\chi_1, \dots, \chi_5$; here we use Mulliken's symbols.

(i) Find the conjugacy classes of Q_8 and their sizes.

(ii) Prove that these five representations are all the irreducible representations of Q_8.

(iii) Write down the character table of Q_8 and compare it with that of D_4.

(iv) Prove that $D_4 \not\cong Q_8$. What does this say about the character table?

Exercise 3. S_4 permutes the coordinates of $\mathbb{C}^4$. Compute the character of this permutation representation, then use the character calculus to decompose it and to prove that its three-dimensional summand is irreducible.

Exercise 4. Let ρ be a d-dimensional representation of a finite group G, and χ its character.

(i) Show that $|\chi(g)| \le d$ for every g, with equality if and only if $\rho(g)$ is a scalar multiple of the identity. Deduce that $\chi(g) = d$ exactly on the kernel of ρ.

(ii) Suppose that ρ is faithful. Prove that $|\chi(g)| = d$ only for central elements g.

(iii) From the tables of Exercise 2, find which irreducible representations of D_4 and Q_8 are faithful, and at which elements $|\chi(g)| = d$.